Exp: 10 **Internetworking with Routers in Cisco Packet Tracer**

Aim:

To design and configure internetworks using routers, a wireless router with DHCP, and a switch-router lab network.

Requirements:

* Cisco Packet Tracer
* Routers, switches, PCs, wireless router, cloud

(a) Simple Internetwork using a Router:

1. Create two networks of 3-4 hosts each, connected through a switch each, and join the switches to a router.
2. Assign static IPs: Network A 192.168.1.0/24 (gw 192.168.1.1), Network B 192.168.2.0/24 (gw 192.168.2.1).
3. Configure router interfaces:

```
Router(config)# interface fa0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config)# interface fa0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
```

4. Ping between hosts of both networks and use Simulation mode to trace how the router forwards the packets.

(b) Wireless Router with DHCP and Internet Cloud:

1. Place a Wireless Router, laptops/smartphones and a Cloud; connect the router's Internet port to the cloud.
2. Open the wireless router GUI > Setup and enable DHCP; set the SSID under Wireless.
3. Configure the wireless laptops to obtain an IP

automatically (DHCP) and connect to the SSID.

4. Verify that clients receive IP addresses and can ping the server through the cloud.

(c) Lab Internetwork using Switch, Router and Ethernet

1. Connect lab PCs to a switch with straight-through cables and the switch to a router interface.
2. Assign IP addresses / gateway and test connectivity using ping.

Result:

Internetworks using routers, a DHCP wireless router and a switch-router lab network were designed and verified.